GRAVITYSORT · RISK REGISTER AND HAZARD ANALYSIS

GravitySort risk register and hazard analysis

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DOCUMENT CONTROL

Document ID	GVS-RSK-001	Version	0.1
Status	Draft	Date	2026-10-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gravitysort		
Source	docs/09-risk.md @ d6d285d		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.8 pipeline): project risk register, preliminary hazard analysis and first-pass design FMEA for the constructable design (option A) at P3

GravitySort risk register and hazard analysis

The top project risk is that the fluidized bowl does not hold fine gold from real feed as well as the paper case assumes, which only a test with characterized ore can settle; the top hazards are the spinning bowl (entanglement, a liner fragment at speed) and a full header tank tipping the machine. Method: STANDARDS section 23 and the risk assessment approach of ISO 12100 (GVS-REG-001 G4). Likelihood, impact, severity, occurrence and detection are scored 1 (low) to 5 (high).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	Fine-gold recovery below PR-004	Bed packs or washes out fine gold; feed rich in heavy minerals saturates the rings before the 2 h flush	Miners see no gain over sluices; adoption fails	3	5	Design lead	Spiked-ore test at TRL 4 before any field trial; adjustable fluidization flow and flush interval
P2	Table cannot reach 100 g from 5.3 kg (PR-005)	Heavy-mineral concentrate, small table	More than 100 g to smelt; borax use and losses rise	3	3	Design lead	Two passes; tune stroke, tilt and bump stop setting at TRL 4
P3	Pedal effort too high at 80 G (PR-006)	Seal drag and slurry power rise with speed squared	Riders tire; motor becomes the default	3	2	Design lead	Run at 60 G; measure union drag; motor option
P4	Over the value-engineering target	Constructable design and fixes add parts ($594 against $455)	Harder for a small group to afford	4	3	Project lead	Savings in GVS-DEC-001 (quarter-turn belt, used bearings, bushes); price check in Colombia
P5	Mass over 100 kg (PR-011)	0.4 kg margin on the estimate; the model's solid volume reads higher	Not carried by two people in six loads	3	3	Design lead	Weigh parts at TRL 4; savings of about 16 kg in GVS-DDR-003 A1 (b)
P6	PU liner cannot be cast reliably or wears fast (PR-010, PR-014)	No local PU casting experience; abrasive quartz	Bowl down time; cost of liners	3	4	Workshop partner	Trial casting at TRL 4; bought liner as fallback
P7	Bought parts not available locally	Rotary union, bevel gearbox, resin	Delay or redesign	3	3	Project lead	Items to confirm list in GVS-DEC-001; quarter-turn belt instead of the gearbox
P8	Partner engagement does not happen	No agreement yet with the cooperative introduced through the Alliance for Responsible Mining	Requirements stay unvalidated	3	4	Project lead (Amish)	Outreach is Amish's action (GVS-DEC-001, 2026-10-02)
P9	Equipment used on mercury-contaminated tailings	Old tailings reprocessed at legacy sites	Mercury spread through the bowl, water and table	2	5	Partner organization	Safety rule and training; test tailings before use

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P10	Concentrate theft	Valuable concentrate at remote sites	Loss to the group; conflict	3	3	Partner organization	Hasps on both containers; two-person rule proposed to the partner

Preliminary hazard analysis

Table 2. Hazards (option A, constructable design).

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Entanglement in belts, chains and pulleys	Unguarded drive, loose clothing or hair	Crushed fingers, scalping	4	2	Belt, chain and table guards fitted before running (PR-012); guard openings to check against ISO 13857 (REG G5)	Low to medium until G5 is checked
H2	Contact with the spinning bowl	Lid opened while the bowl coasts (about 20 s)	Hand injury	4	2	Lid interlock pin on the brake cable; disc brake with parking latch stops it in 0.6 s	Low
H3	Liner or shell fragment at speed	Debonded or cracked liner; over-speed by a sprinting rider	Impact injury	5	1	Burst safety factor 16 at 1,200 rpm; liner check before every run; speed display; containment shown face first only (PR-016)	Medium until the impact test or the band decision (GVS-DEC-001 O2)
H4	Machine tips with a full tank	Uneven ground, side push on the tank, climbing on the frame	Crush injury	4	2	Tank on the centre line, braced post; stands on 13.1 degrees (PR-015); fill only standing level	Low
H5	Table head and pitman	Hands near the eccentric or bump stop while running	Pinched fingers	3	2	Table belt guard; stop adjusted only with the belt slack	Low
H6	Slips and drowning	Slurry underfoot; open settling ponds and pits	Falls; drowning	5	2	Boots with grip; fence ponds (site rule with the partner)	Medium (site dependent)
H7	Mosquito breeding in ponds	Standing water in settling ponds and pits	Malaria and other vector-borne disease	4	3	Pond management with the partner (cover, drain, recirculate); STM S30	Medium (site dependent)
H8	Arsenic and lead in concentrate	Arsenopyrite and galena in heavy-mineral concentrate	Poisoning by ingestion	4	2	Gloves, hand washing, no eating at clean-up (STM S31)	Low to medium
H9	Mercury cross-contamination	Processing old amalgamation tailings	Mercury exposure and spread	5	2	Never use with mercury or	Low with the rule kept

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
						untested tailings (STM S28)	
H10	Smelting heat and borax	Smelting outside the machine at about 1,065 C; borax is Repr. 1B (REG G10)	Burns; reproductive toxicity	4	2	Furnace, tongs, face shield, ventilation, outdoors; partner's smelting procedure	Medium (outside this design)
H11	Battery and electrical (motor option)	Lithium pack, controller	Fire, shock	4	1	MotionCore safety section; hardwired e-stop; mechanical brake still used	Low
H13	Full tank sways when the table runs	Post above the brace node about 3.5 Hz, table drive 4.0 to 5.0 Hz with impact harmonics (STEM S33)	Weld fatigue, tank falling, tipping	4	2	Interim rule: run the table with the drum drained; raise the brace node (GVS-DEC-001 O3)	Medium until O3 is decided
H12	Manual handling	Loads up to 27.3 kg carried to site	Back injury	3	3	Two-person carries, loads 30 kg or less (PR-011)	Medium

Design FMEA (first pass, P3)

Table 3. Failure modes, option A. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Bowl liner (BOM 3)	Hold the bed in the riffle rings	Debonds from the shell or cracks	Fragment risk; bed lost	Poor casting, wear, impact	5	3	3	45	Visual and tap check before every run; impact test at TRL 4; band option (O2)
F2	Fluidization holes (BOM 3)	Keep the bed loose	Holes block, or flow too low to loosen a coarse bed	Bed packs; recovery falls	Silt and sand in recirculated water; 2.7 mm/s upflow near the minimum fluidization of 75 um grains	4	3	3	36	Screen filter fitted (F2 fix); flush the filter when the rotameter drops; hole check at each flush
F3	Rotary union (BOM 4)	Pass water into the turning spindle	Leaks or seal drag rises	Water loss, more pedal power	Silt in the seal, wear	3	3	2	18	Filter upstream; datasheet seal drag; spare seal kit
F4	Header supply (BOM 13)	Give 8.1 kPa at the union	Tank runs low	Fluidization weak at the lower rings	Upstream supply interrupted	3	3	2	18	Keep the tank above half full; rotameter visible to the operator
F5	Disc brake and interlock (BOM 18)	Stop the bowl; lock the lid	Cable stretches; pin misses	Lid opens while coasting	Wear, poor adjustment	4	2	3	24	Check the interlock at each flush; adjust the cable
F6	V-belts (BOM 10)	Drive the bowl and table	Slip or break	Bowl slows; G falls	Wear, wet belts	2	3	2	12	Speed display shows it; spare belts
F7	Spindle bearings (BOM 5)	Carry the spindle	Wear, seizing	Vibration, stop	Slurry ingress	3	2	3	18	Standpipe shields the upper bearing; grease interval at TRL 4
F8						3	3	2	18	Buffer rated 800 N (F3

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
	Table bump stop (BOM 22)	Make the stroke asymmetric	Buffer tears or loosens	Gold stops walking	Fatigue at 270 strokes/min					fix); lock nuts; spare buffer
F9	Tank post and braces (BOM 1)	Hold the full tank 1.7 m up	Weld cracks	Tank falls; tipping	Fatigue from sway near the table drive frequency (about 3.5 Hz against 4 to 5 Hz) and knocks	4	3	3	36	Raise the brace node (O3); drain the drum before running the table; weld inspection
F10	Pedal drive (BOM 8, 9)	Turn the jackshaft	Chain off, freewheel slips	Stop; no harm	Wear, misalignment	2	3	1	6	Chain case; tension check
F11	Frame members (BOM 1)	Carry bearings and loads	Bearing misalignment; first critical speed lowered by member compliance (1,274 to 1,976 rpm)	Vibration, belt wear; resonance near a sprint speed	Spindle member deflection 0.73 mm	4	2	3	24	Tap test and alignment check on the first frame (O4); never run above 900 rpm
F12	Concentrate box and flush container hasps (BOM 16, 23)	Secure the concentrate	Forced open	Theft	Weak hasp	3	2	2	12	Two-person rule proposed to the partner